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The sterile insect technique: an international framework to facilitate transboundary shipments of sterile insects

W.R. Enkerlin* & R. Pereira

Insect Pest Control Subprogramme, Joint FAO/IAEA Centre of Nuclear Techniques in Food and Agriculture, Department of Nuclear Sciences and Applications, International Atomic Energy Agency, PO Box 100, A-1400 Vienna, Austria

*Corresponding author: w.r.enkerlin@iaea.org

Summary

The sterile insect technique (SIT) has been successfully used since the 1950s as part of an integrated pest management approach in large-scale programmes to prevent, contain, suppress and eradicate key insect pests in many countries throughout the world. During this period, over one trillion live sterile insects have been shipped across borders. The very few adverse incidents from this significant trade were managed and resulted in no significant impacts. The phytosanitary and zoosanitary requirements by importing countries have been simple, facilitating the transboundary shipment of sterile insects, which is carried out mostly under the framework of cooperative agreements between the governments of the countries involved, and under technical cooperation projects of the United Nations.

However, the shipment of sterile insects from sources outside this governmental framework, including public–private facilities, has been complicated, despite the availability of harmonised international guidelines in some cases, such as those for fruit flies. The SIT has great potential for the control of endemic pests or against the growing threat of invasive pests that can affect whole regions and even continents. Since SIT is species-specific, with negligible risk of introducing unwanted invasive species to the environment, and with the advantage of reducing insecticide use, a harmonised framework that recognises the low risk of SIT would facilitate shipments of sterile insects across borders and help to expand the use of this effective and environmentally friendly technology. The scope of this paper is limited to insects that have been sterilised using ionising radiation.

Keywords

Harmonised guidelines – Ionising radiation – Sterile insect technique – Sterile insects – Transboundary shipments.

Introduction

The sterile insect technique (SIT) has been successfully used in large-scale programmes to prevent, contain, suppress and eradicate pests in many countries throughout the world [1]. Since the SIT is species-specific, it enables pest control without the risk of introducing potentially invasive insect species into ecosystems, threatening their biodiversity. Unlike insecticides and other control methods, its efficiency increases as the population density of the target pest decreases [2].

The SIT described in this article specifically uses ionising radiation – gamma rays, X-rays or electron-beam (e-beam) processing – to sterilise mass-reared insects of the target pest population so that they cannot produce offspring when they are released in the field and mate with their wild counterparts. As a result of the physical action of radiation, chromosomes break in the germinative cells of the insects' sperm and ovules, causing lethal dominant mutations that prevent the development of the embryos [3]. (Although SIT can be applied to both males and females, releasing only sterile males is more efficient.) This results in no offspring in the wild population and the reduction of the pest population over time.

This method relies on the sustained and area-wide release of large numbers of sterile insects over areas infested with the target pest. Before release, sterile insects are mass-reared, sterilised, packed, shipped, have emerged as adults from the puparia, been fed and matured, and are then loaded into delivery vehicles for aerial or ground release

Figure 1

Metal box containing sterile fruit flies, being loaded onto an airplane for aerial release

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(Figure 1). The conditions under which these activities take place are as relevant to the overall success of – and negligible risk from – SIT activities as the production of high-quality sterile insects [4].

One such pest prevention operation is the Mediterranean Fruit Fly Preventive Release Program in California, the United States of America (USA), in which sterile Mediterranean fruit flies or 'medflies' (*Ceratitis capitata*) are released weekly, by air, over large areas at risk of introductions of this invasive pest [5]. Sterile insects have also been used in biological containment barriers, such as the medfly barrier located along the Mexico–Guatemala border. Over one billion sterile medflies are released along this border every week to prevent their spread to the medfly-free areas north of the barrier [6].

In the Okanagan–Kootenay Sterile Insect Release (OKSIR) programme to suppress codling moth (*Cydia pomonella*) in Okanagan, British Columbia, Canada, sterile moths are continuously released over apple orchards to control the pest, as a safe replacement for organophosphate insecticides [7].

To eradicate the New World screwworm (*Cochliomyia hominivorax*) from Central America, sterile *Cochliomyia* were reared and released throughout the whole region [8]. The SIT method has also been used to eradicate tsetse fly (*Glossina austeni*) populations on Unguja Island, Zanzibar, between 1994 and 1997. This was the first SIT-based eradication effort against a tsetse species [9]. An operation to eradicate *Glossina palpalis gambiensis* in Senegal is ongoing [10]. The SIT approach is also being used to eradicate incipient outbreaks of invasive pest species, such as the cactus moth (*Cactoblastis cactorum*) in the Yucatan Peninsula, Mexico [11]; a major outbreak of the New World screwworm in Libya [12] and medfly in the Dominican Republic [13].

Benefits and risks of the sterile insect technique

The return on investment provided by past and current SIT programmes is unquestionable, with benefit-to-cost ratios ranging from US$ 2.80 for each dollar invested in the case of a medfly population suppression programme in South Africa, to as much as 1,000 to 1 in the case of a medfly prevention programme in Chile [14]. The SIT has been applied for more than 65 years against several insects which are key plant pests, such as fruit flies and moths, and key livestock pests, some of which are vectors of zoonotic diseases, including New World screwworm and tsetse flies. More recently, SIT has been developed against *Aedes aegypti* and *Aedes albopictus* mosquitoes and vectors of human diseases such as dengue, chikungunya and Zika. Pilot trials are under way in several countries, including Brazil, the People's Republic of China, Cuba, Germany, Italy, Mauritius, Mexico, Singapore, Spain, and the USA. These pilot trials have clearly demonstrated the impact of releases of sterile male mosquitoes over wild populations, and a corresponding reduction in the transmission and incidence of disease in the human population [15].

Sterile insects used in SIT programmes have been successfully shipped across borders to many countries since 1963. Although the 58 years of transboundary shipments have not been well documented, it is known that very few incidents have occurred. Such incidents have mainly been related to production protocols, demonstrating the need for quality assurance in the sterilisation process and highlighting the advantages of well-resourced, centralised production facilities. For example, one case involved an error in the irradiation of New World screwworm that resulted in the release of fertile flies. In that instance, the releases were near the production facility and the fertile flies were contained and eliminated in a short period of time, with no significant impacts on livestock or human health. Another case involved the escape of fertile medflies from a production unit. The flies were detected through a trapping network and contained around the facility by applying a rapid response protocol, with no further consequences.

Other issues have arisen from delays in international transport, resulting in the release of poor-quality flies. In one instance, labelled boxes with sterile medfly pupae were held at Customs during transit, causing an excessive period of anoxia before the pupae arrived at their destination. This was due to inadequate documentation causing confusion at Customs.

Production and sterilisation processes

Proper production conditions are key to virtually eliminating risk from large shipments of sterile insects. Producing sterile insects for use in SIT programmes involves mass

rearing with high quality standards. Mass-rearing facilities employ standard operating procedures, set out in manuals that describe the production process in detail. The process involves breeding colonies; egg seeding; rearing larvae, pupae and adults; and irradiation. Generally, each of these steps requires separate rearing areas and specific climatic conditions, including temperature and relative humidity. The diet used to feed adults in the breeding colonies and the one used to rear the larvae are specific to each insect species. During the rearing of the different biological stages of the insects, their diets are kept free of pathogenic microorganisms by manipulating the pH, applying specific antibiotics and fungicides and, in certain cases, through sterilisation. To maintain high levels of hygiene inside the facility and prevent contamination of the diets and rearing environment, specific biosecurity protocols are followed at all times [16].

Sterilisation of pupae or adults also follows specific irradiation protocols, aimed at inducing the required sterility without compromising the quality of the irradiated insects. Before irradiation, insects are marked with a specific fluorescent dye that enables the released sterile insects to be differentiated from wild insects when they are caught in traps [17]. Biosecurity protocols are in place to allow traceback of each batch of irradiated insects in case there is any need to check for a faulty procedure [4]. The possibility of any introduced pathogen, parasite, or other form of contaminant in the production process being undetected is extremely low.

The SIT programmes rely on the ability to measure and control the irradiation dose delivered to the insects for their sterilisation. This is essential to balance effective sterilisation with the utility of the intervention in terms of the quality and vitality of the treated insects. Programmes have well-established dosimetry systems to accurately measure the absorbed dose and estimate if the associated confidence interval is within the permissible range. This applies to any insect pest of agricultural, veterinary, and medical importance subject to sterilisation through ionising radiation. Procedures are available, such as dose mapping by scanning Gafchromic® film, to measure the radiation dose absorbed by the insects during their sterilisation [18]. This practically eliminates the risk of accidental introductions of insects that can establish and persist over generations.

Specific and rigorous quality controls are applied to the mass-rearing and sterilisation processes and to the insect itself. The aim is to ensure the high quality of sterile released insects. Standardised quality control procedure manuals are available for sterile insects used in operational programmes, such as the one for Tephritid fruit flies [19]. In this way, the risk to the utility of live insects is managed.

Post-production process

As in the case of the production process, the post-production handling of sterile insects is a delicate and rigorous process. Standard operating procedures are available to ensure proper handling and good sterile insect quality. Strict biosecurity protocols are in place in the emergence and release facility, as well as for long-distance transportation, to ensure the integrity of the consignment and prevent the accidental spread of unwanted insects (hitchhikers) (Figure 2) [16, 20].

The need for international shipments in sterile insect release programmes

The production and sterilisation of sterile insects in centralised mass-rearing facilities is a common practice [21]. Centralised production and supply of sterile insects provides many advantages, such as more effective mass-rearing and sterilisation processes, more uniform quality of the sterile insects, and a lower cost per million insects due to economies of scale [22]. Nevertheless, in certain situations, smaller local facilities ('the right size facility') may offer more flexibility, provide certainty in ensuring the local availability of sterile insects, and provide resilience against factors such as problems with shipping. Sterile insects from centralised facilities can be used for field releases on site and/or to supply SIT programmes abroad, through transboundary and sometimes long-distance shipments. The largest insect mass-rearing facilities in the world include the El Pino facility in Guatemala, with a production capacity of 2.5 billion sterile insects per week; the newly inaugurated facility in Metapa, Mexico, with a production capacity of one billion sterile insects per week; and a facility in Valencia, Spain, with a production capacity of 700 million sterile insects per week. All three facilities produce sterile medflies. These and other facilities have been a source of sterile fruit flies for many years, shipping live insects as eggs, pupae and adults across international borders to be used in SIT programmes in other countries [21].

Figure 2

Boxes containing sterile insects properly labelled for shipping

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Since the mid-1950s, when the use of SIT began in operational programmes, more than one trillion sterile insects have been shipped across borders to supply programmes in 31 recipient countries from 16 sterile insect production facilities, based in 11 countries (see the World-Wide Directory of SIT Facilities, DIR–SIT, at https://nucleus.iaea.org/sites/naipc/dirsit/PublicPages/Home.aspx). Included on the list at this database are shipments of large volumes of the New World screwworm (*C. hominivorax*), the medfly (*C. capitata*), the Mexican fruit fly (*Anastrepha ludens*), the tsetse fly (*Glossina* spp.), pink bollworm (*Pectiniphora gossypiella*), and smaller shipments of other insect species, such as codling moth (*C. pomonella*), and cactus moth (*C. cactorum*), listed in Table I. This has mostly been carried out under a framework of cooperative agreements between countries and under technical cooperation projects of the United Nations.

Nevertheless, in recent years, with the increasing interest in the use of SIT against major insect pests and the involvement of public–private facilities in mass insect rearing and sterilisation, transboundary shipments of sterile insects outside the government programme framework have been conducted mainly for pilot trials. These shipments have faced significant issues with the plant regulatory authorities in some countries, as well as with transportation companies, which are confronted with a new and uncertain situation. This has significantly delayed some shipments and even prevented others altogether.

Harmonised guidelines adopted by international conventions, like those of the International Plant Protection Convention (IPPC) and World Organisation for Animal Health (WOAH, founded as OIE), would serve as a framework for both government and non-governmental organisations and agencies, and would greatly facilitate transboundary shipments of sterile insects. If such guidelines were developed, they would significantly contribute towards fostering the expansion of SIT to more countries and more insect pests.

Table I

History of transboundary shipment of sterile insects until 2021

Year	Species	Site of production	Approximate amount shipped (million pupae)	Recipient	Purpose
1963–2000	Mexican fruit fly, *Anastrepha ludens*	Mexico (Monterrey and Metapa)	2,511.00	USA (Texas and San Diego)	To assist the California and Texas Departments of Agriculture in the eradication of Mexican fruit fly outbreaks
1970–2021	Mediterranean fruit fly, *Ceratitis capitata*	Argentina (Mendoza), Austria (Seibersdorf), Costa Rica, Guatemala (El Pino), Israel (Biofly), Mexico (Metapa), Portugal (Madeira), Spain (Valencia)	911,536.00	Argentina (Patagonia), Bolivia (Cochabamba), Chile, Croatia, Ecuador, Italy (Procida), Greece, Guatemala, Honduras, Israel, Jordan, Mexico, Morocco, Nicaragua, Spain (Canary Islands), Tunisia, USA	For prevention, containment, suppression, and eradication of the Mediterranean fruit fly
1990–1991	New World screwworm (NWS) *Cochliomyia hominivorax*	Austria (Seibersdorf), Mexico (Tuxtla Gutierrez, Chiapas), Panama (Panama–USA Commission for the Eradication and Prevention of Screwworm or COPEG)	108,994.00	Central American countries (7 countries), Libya (Tripoli), USA (Florida)	To eradicate NWS from Central America To assist the eradication of NWS outbreaks in Libya and Florida, USA
1990–2021	Tsetse fly (*Glossina* spp.)	Austria (Seibersdorf), Slovakia	25.50	Tanzania (Tanga) Senegal (Dakar)	To assist tsetse eradication from Tanzania and Senegal
2008	Cactus moth (*Cactoblastis cactorum*)	USA (Florida)	0.02	Mexico (Yucatan Peninsula)	Eradication of incipient cactus moth outbreaks
2001–2014	Pink bollworm (*Pectinophora gossypiella*)	USA (Arizona)	Unknown	Mexico (Northern States)	Regional (USA and Mexico) eradication of the pink bollworm
2010–2020	Codling moth (*Cydia pomonella*)	Canada (Osoyoos)	Unknown	New Zealand, South Africa, USA	For sterile insect technique pilot validation
	Total		**1,023,066.50**		

Source: updated from FAO/IAEA/USDA statistics [19]

Regulations

Sterile insect technology has been recognised by the IPPC as a phytosanitary measure for pest control. It is defined as a: 'Method of pest control using area-wide inundative release of sterile insects to reduce reproduction in a field population of the same species' [20]. Sterile insects are defined as: 'An insect that, as a result of a specific treatment, is unable to reproduce' [20].

Moreover, in its 'Guidelines for the export, shipment, import and release of biological control agents and other beneficial organisms' [20], the Food and Agriculture Organization of the United Nations (FAO) makes a distinction between biological control agents capable of self-replication (including parasitoids, predators, parasites, nematodes, phytophagous organisms, and pathogens such as fungi, bacteria, and viruses), and insects and other beneficial organisms that are sterile. The fact that sterile insects are not capable of self-replicating is a major advantage in protecting the environment, compared with other insects that are shipped across borders for biological control or other purposes.

A hazard analysis was conducted in 2001 by an FAO/International Atomic Energy Agency (IAEA) Consultants Group to determine the level of risk of potential hazards from the transboundary shipment of sterile insects. These included:

- outbreaks of the target insect pest in a new area
- increasing the fitness of the local pest population
- the initiation of unnecessary regulatory action
- the introduction of exotic (new) contaminant organisms.

The study concluded that the present systems of transboundary shipment of sterile insects for SIT programmes are very safe. Nevertheless, the study recommended that a harmonised guideline in the form of an international standard should be developed for approval by the IPPC Commission on Phytosanitary Measures, to facilitate commercial development of the SIT [4].

For insects that may be plant pests, the National Plant Protection Organization (NPPO) of each country should designate the proper authority to ensure safe shipment of sterile insects (either through or to their territory). It is up to the NPPO to coordinate with the producer/shipper on their responsibilities for achieving safe shipment, because producers of sterile insects may be governmental organisations or private businesses, parastatal (a state-owned enterprise), part of a joint venture or an internationally owned facility.

Historically, the transboundary shipment of sterile insects has been controlled by simple basic phytosanitary regulations and documentation (Figure 3). These include:

- an import permit
- a national transit permit
- a phytosanitary certificate
- an irradiation certificate
- labelling and notification
- transit regulations, if the shipment travels through a third country (i.e. neither the country of origin nor the country of destination of the consignment).

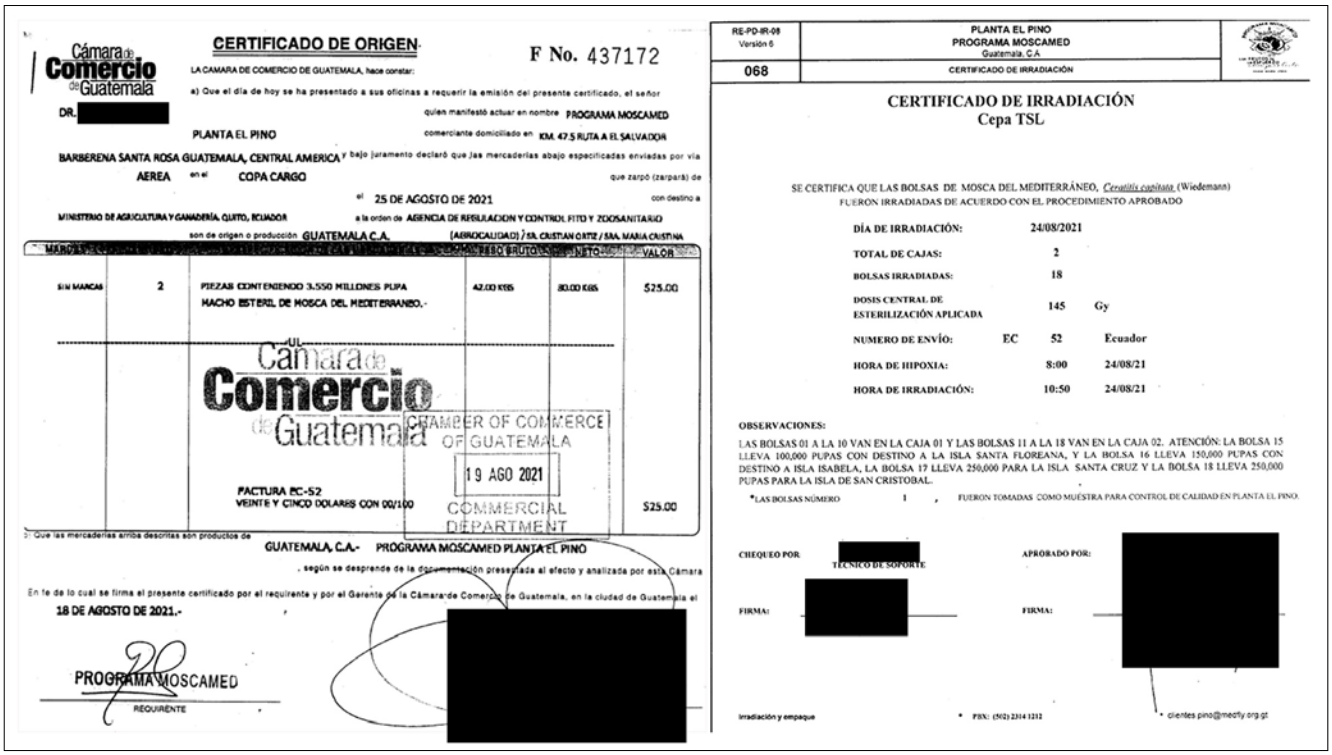
The International Standard for Phytosanitary Measures (ISPM) No. 3: 'Guidelines for the export, shipment, import and release of biological control agents and other beneficial organisms' states, in its section on 'Responsibility of Importers', that importers should ensure that organisms for SIT have been treated to achieve the required sterility for SIT purposes (e.g. using irradiation with the required minimum absorbed dose). The treatment(s) used and an indication of the effectiveness of the sterilisation process should also be provided [20].

The FAO/IAEA manual entitled *Guideline for Packing, Shipping, Holding and Release of Sterile Flies in Area-Wide Fruit Fly Control Programmes* [24] includes a chapter describing the regulatory procedures involved in transboundary shipments of sterile insects. These procedures include the responsibilities of the producer/shipper of sterile insects, of the authorities before export and upon import (final or in transit) and of the importer [6]. No shipment of sterile insects has ever been rejected by national or international plant protection or regulatory authorities [23].

Future perspectives

Future investment in SIT may focus on large, centralised sterile insect production facilities from which insects can be shipped abroad to supply the needs of countries who want to apply the technology. This would avoid, when appropriate, the need for each country to build its own smaller facilities [21, 24]. The advantages of such a scheme are twofold: it would reduce the risk of non-native insects escaping from in-country breeding colonies before sterilisation; and it would enable greater quality control of the sterile insects, a vital aspect of the effective application of SIT. Sterile insects would be available at more competitive prices, as insect mass rearing is susceptible to economies of scale [22]. Nevertheless, an alternative scheme, where smaller 'right-size' facilities are built in user countries, based on local needs, could provide greater flexibility and certainty in securing the required numbers of sterile insects, especially if there are uncertainties about the reliability of international shipments. This approach would require significant capital investment for each facility.

In order to take advantage of larger centralised facilities, an international framework is needed for transboundary shipments and the importation (whether in transit or as the final



CERTIFICADO DE ORIGEN

F No. 437172

Cámara de Comercio de Guatemala

LA CAMARA DE COMERCIO DE GUATEMALA, hace constar:

a) Que el día de hoy se ha presentado a sus oficinas a requerir la emisión del presente certificado, el señor DR. [redacted] quien manifestó actuar en nombre PROGRAMA MOSCAMED PLANTA EL PINO comerciante domiciliado en KM. 47.5 RUTA A EL SALVADOR BARBERENA SANTA ROSA GUATEMALA, CENTRAL AMERICA y bajo juramento declaró que las mercaderías abajo especificadas enviadas por vía AEREA en el COPA CARGO que zarpó (zarpará) de el 25 DE AGOSTO DE 2021 con destino a MINISTERIO DE AGRICULTURA Y GANADERÍA, QUITO, ECUADOR a la orden de AGENCIA DE REGULACION Y CONTROL FITO Y ZOOSANITARIO son de origen o producción GUATEMALA C.A. (AEROCALIDAD) / SRA. CRISTIAN ORTIZ / SRA. MARIA CRISTINA

MARCA			PESO BRUTO	PESO NETO	VALOR
SIN MARCAS	2	PIEZAS CONTENIENDO 3,550 MILLONES PUPA MACHO ESTERIL DE MOSCA DEL MEDITERRANEO.-	42.00 KGS	80.00 KGS	$25.00
		FACTURA EC-52 VEINTE Y CINCO DOLARES CON 00/100			$25.00

Cámara de Comercio de Guatemala CHAMBER OF COMMERCE OF GUATEMALA 19 AGO 2021 COMMERCIAL DEPARTMENT

3.- Que las mercaderías arriba descritas son productos de GUATEMALA, C.A.- PROGRAMA MOSCAMED PLANTA EL PINO, según se desprende de la documentación presentada al efecto y analizada por esta Cámara

En fe de lo cual se firma el presente certificado por el requirente y por el Gerente de la Cámara de Comercio de Guatemala, en la ciudad de Guatemala el 18 DE AGOSTO DE 2021.-

PROGRAMA MOSCAMED

REQUIRENTE

[redacted]

RE-PD-IR-08 Versión 6 | PLANTA EL PINO PROGRAMA MOSCAMED Guatemala, C.A. | 068 | CERTIFICADO DE IRRADIACIÓN

CERTIFICADO DE IRRADIACIÓN
Cepa TSL

SE CERTIFICA QUE LAS BOLSAS DE MOSCA DEL MEDITERRÁNEO, *Ceratitis capitata* (Wiedemann) FUERON IRRADIADAS DE ACUERDO CON EL PROCEDIMIENTO APROBADO

DÍA DE IRRADIACIÓN:	24/08/2021		
TOTAL DE CAJAS:	2		
BOLSAS IRRADIADAS:	18		
DOSIS CENTRAL DE ESTERILIZACIÓN APLICADA	145	Gy	
NUMERO DE ENVÍO:	EC	52	Ecuador
HORA DE HIPOXIA:	8:00	24/08/21	
HORA DE IRRADIACIÓN:	10:50	24/08/21	

OBSERVACIONES:

LAS BOLSAS 01 A LA 10 VAN EN LA CAJA 01 Y LAS BOLSAS 11 A LA 18 VAN EN LA CAJA 02. ATENCIÓN: LA BOLSA 15 LLEVA 100,000 PUPAS CON DESTINO A LA ISLA SANTA FLOREANA, Y LA BOLSA 16 LLEVA 150,000 PUPAS CON DESTINO A ISLA ISABELA, LA BOLSA 17 LLEVA 250,000 PARA LA ISLA SANTA CRUZ Y LA BOLSA 18 LLEVA 250,000 PUPAS PARA LA ISLA DE SAN CRISTOBAL.

*LAS BOLSAS NÚMERO 1 , FUERON TOMADAS COMO MUESTRA PARA CONTROL DE CALIDAD EN PLANTA EL PINO.

CHEQUEO POR: [redacted] TÉCNICO DE SOPORTE

APROBADO POR: [redacted]

FIRMA: [redacted]

FIRMA:

Irradiación y empaque * PBX: (502) 2314 1212 * clientes.pino@medfly.org.gt

Figure 3

Official documents used for transboundary shipment of sterile insects (certificate of origin and radiation certificate)

Signatures and names have been redacted for data-protection purposes

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destination) of sterile insects for use in control programmes. Such a framework should recognise the extensive record of quality control and safe shipments of mass-reared insects sterilised through ionising radiation. This would serve as the basis for more harmonised regional or national phytosanitary and animal health measures [23] and should apply to both government and private entities that follow the same procedures.

Conclusions

Sterile insect technology is a cost-effective and environmentally friendly pest-control tool that has been extensively applied to control a range of major insect pests. Shipments of sterile insects across borders have taken place for more than 65 years, requiring only basic phytosanitary regulations.

Risks from the production phase are managed through quality control and agreed standard operating procedures that cover the key steps. The fact that these insects are sterile essentially eliminates concerns around the introduction of invasive or exotic species. No major incidents associated with transboundary shipments of sterile insects have occurred since this activity began in the 1950s.

Taking into account the negligible risk and animal health demonstrated through many years of use, phytosanitary regulations for transboundary shipments of sterile insects should be kept as simple as possible. A harmonised framework to facilitate shipments of sterile insects across borders, regardless of the parties involved (public or private), would foster the use of this effective and environmentally friendly technology.

Acknowledgements

The authors would like to acknowledge the managers of sterile insect production facilities for providing data on the numbers of sterile insects shipped across borders to support pest-control programmes in many countries.

La technique de l'insecte stérile : un cadre international pour faciliter les transports transfrontaliers d'insectes stériles

W.R. Enkerlin & R. Pereira

Résumé

La technique de l'insecte stérile (TIS) est utilisée avec succès depuis les années 1950, en tant que composante d'une approche intégrée de gestion des ravageurs mise en place par des programmes de grande envergure appliqués dans de nombreux pays pour prévenir, maîtriser, supprimer et éliminer les principales espèces d'insectes nuisibles. Depuis lors, plus d'un trillion d'insectes stériles vivants ont été transportés au-delà des frontières nationales. Les très rares incidents indésirables survenus en marge de ces volumineux échanges ont été maîtrisés et n'ont eu aucun impact significatif. La simplicité des exigences phytosanitaires et de santé animale requises par les pays importateurs a facilité le transport transfrontalier des insectes stériles, le plus souvent effectué dans le cadre d'accords de coopération conclus entre les gouvernements des pays concernés et de projets de coopération technique des Nations Unies.

En revanche, il est plus compliqué de transporter des insectes stériles issus de sources extérieures aux cadres gouvernementaux, en particulier des sites relevant d'une gestion publique-privée, et ce malgré l'existence de directives internationales harmonisées, par exemple celles s'appliquant aux différentes mouches de fruits. La TIS présente un grand potentiel pour lutter contre les ravageurs endémiques ou contre la menace croissante associée aux ravageurs envahissants, qui peuvent affecter des régions entières, voire des continents. Étant donné que la TIS est spécifique pour chaque espèce, qu'elle présente un risque négligeable d'introduire des espèces envahissantes non désirées et qu'elle offre l'avantage de réduire le recours aux insecticides, il conviendrait de disposer d'un cadre harmonisé qui reconnaisse le faible niveau de risque associé à la TIS et facilite ainsi les transports transfrontaliers d'insectes stériles, afin d'aider au déploiement plus large de cette technologie efficace et sans risque pour l'environnement. La portée de cet article se limite aux insectes stérilisés par une exposition aux radiations ionisantes.

Mots-clés

Directives harmonisées – Insectes stériles – Radiation ionisante – Technique de l'insecte stérile – Transports transfrontaliers.

Técnica del insecto estéril y ordenamiento internacional para facilitar los envíos transfronterizos de insectos estériles

W.R. Enkerlin & R. Pereira

Resumen

Desde el decenio de 1950 se viene empleando con éxito la técnica del insecto estéril (*sterile insect technique*, SIT), aplicada en el manejo integrado de plagas, como parte de programas que se despliegan a gran escala para prevenir, contener, suprimir y erradicar importantes plagas de insectos en muchos países del mundo. En todos estos años, más de un billón de insectos estériles vivos han sido transportados en cargamentos transfronterizos. Los contados incidentes adversos ocasionados por este voluminoso comercio fueron debidamente manejados y se saldaron sin repercusiones de importancia. Los requisitos fitosanitarios y zoosanitarios impuestos por los países importadores han sido sencillos, cosa que ha facilitado el envío transfronterizo de insectos estériles, que mayoritariamente se inscribe en acuerdos de cooperación suscritos entre los gobiernos de los países en cuestión y en proyectos de cooperación técnica de las Naciones Unidas.

Sin embargo, el transporte de insectos estériles procedentes de fuentes no sujetas a estos regímenes oficiales, incluidas instalaciones publico-privadas, ha sido complicado, pese a la existencia de directrices internacionales

armonizadas aplicables en ciertos casos, como el de la mosca de la fruta. La técnica del insecto estéril puede ser sumamente útil para luchar contra plagas endémicas o contra la creciente amenaza de plagas invasoras capaces de afectar a regiones enteras o incluso a continentes enteros. Teniendo en cuenta que esta técnica utiliza en cada caso la liberación de insectos estériles para controlar insectos plaga de la misma especie, que entraña por ello un riesgo insignificante de introducir en el medio especies invasoras indeseadas y que además presenta la ventaja de reducir el uso de insecticidas, la existencia de un ordenamiento armonizado que reconozca el escaso nivel de riesgo que presenta la SIT facilitaría el envío transfronterizo de insectos estériles y ayudaría a extender el uso de esta técnica tan eficaz como ambientalmente respetuosa. Las consideraciones que aquí hacen los autores se aplican únicamente al caso de los insectos esterilizados con radiación ionizante.

Palabras clave

Directrices armonizadas – Envíos transfronterizos – Insectos estériles – Radiación ionizante – Técnica del insecto estéril.

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